

UKG 17-Axis System Configuration Files

Below are the YAML configuration files that define the complete UKG/USKD 17-axis system. The configuration is organized into a master workflow and multiple subsystem modules, following a clear hierarchy. Each YAML uses structured keys, indentation, and numbered nodes for readability, with explanatory notes to aid interpretability. The master workflow orchestrates all subsystems, while each subsystem YAML (17-axis schema, 10-layer simulation stack, 12-step refinement engine, quad personas, and KA registry) can plug into the master for unified operation.

Master Workflow YAML

This master workflow YAML coordinates the full pipeline. It references all subsystems, initializes the environment, and defines the execution sequence from input parsing through simulation and refinement, including dynamic control-flow. Explanation nodes (e.g. `description` and `details`) clarify each step. Triggers and conditions are specified to handle persona arbitration on conflicts, compliance rollbacks, iterative refinement loops, and final output assembly.

```
master_workflow:
  subsystems:
    axis_schema_config: "17_axis_coordinate_schema.yaml"
    simulation_stack_config: "10_layer_simulation_stack.yaml"
    refinement_engine_config: "12_step_refinement_engine.yaml"
    persona_templates: "quad_persona_templates.yaml"
    knowledge_agent_registry: "ukg_knowledge_agents_registry.yaml"
  initialization:
    load_all_subsystems: true
    description: "Initialize knowledge graph and all subsystems for new query"
  execution_flow:
    step_1:
      operation: "Parse Input Coordinate"
      description: "Convert user query into 17-axis coordinate values
(Temporal, Spatial, etc.)"
      invoke: axis_schema_config
    step_2:
      operation: "Activate Personas"
      description: "Spawn four AI personas (Knowledge, Sector, Regulatory,
Compliance) with context"
      invoke: persona_templates
    step_3:
      operation: "Run Simulation Stack"
      description: "Execute the 10-layer recursive reasoning process across
personas"
```

```

    invoke: simulation_stack_config
  step_4:
    operation: "Run Refinement Engine"
    description: "Perform 12-step iterative refinement on the simulation
output"
    invoke: refinement_engine_config
  step_5:
    operation: "Output Finalization"
    description: "Compile and format the final answer after all checks"
    invoke: "Final Output Module"
  triggers:
    - condition: "persona_conflict_detected"
      action: "persona_arbitration"
      details: "If personas disagree, initiate consensus debate (Layer 5
coordination) to resolve conflicts 1 2 "
    - condition: "compliance_violation_detected"
      action: "containment_rollback"
      details: "If a Regulatory/Compliance rule is violated, rollback to a safe
state and adjust plan to satisfy constraints"
    - condition: "refinement_loop_incomplete"
      action: "iteration_feedback"
      details: "If confidence < threshold after refinement, feed insights back
and repeat refinement (up to max iterations)"
    - condition: "threshold_not_met_after_max"
      action: "manual_review_alert"
      details: "If output still low-confidence after max loops, flag for human
review before finalizing"
    - condition: "all_clear"
      action: "finalize_output"
      details: "All checks passed and confidence threshold met - produce final
answer and terminate workflow"

```

(The master workflow orchestrates query processing through all subsystems, handling dynamic decisions such as persona consensus and iterative refinement.)

17-Axis Coordinate Schema YAML

This YAML defines the 17-axis coordinate schema used by the UKG system 3 . Each axis is listed with a number and name, along with a brief description of its dimension of knowledge. The schema supports parsing high-dimensional coordinates by mapping input data into these axes. This ensures that any scenario or knowledge node can be encoded along all relevant dimensions. An example mapping is included to illustrate how a complex scenario might populate values on each axis.

```

axes:
  "1. Temporal Axis":
    description: "Time context (when events occur; dates, times, sequences)"

```

"2. Spatial Axis":
 description: "Geographic/Location context (where events occur; locations, places)"

"3. Semantic Axis":
 description: "Conceptual/Ontological context (definitions, concepts, relationships)"

"4. Procedural Axis":
 description: "Process & Methods context (how things are done; workflows, steps)"

"5. Analytical Axis":
 description: "Quantitative analysis & models (what data analysis shows; statistics, predictions)"

"6. Normative Axis":
 description: "Policies, rules & standards (what should be done; guidelines, best practices)"

"7. Risk/Impact Axis":
 description: "Risks, threats & consequences (what could go wrong or right; risk assessments)"

"8. Performance Axis":
 description: "Performance & efficiency metrics (how well things work; outcomes, KPIs)"

"9. Expertise Axis":
 description: "Expert knowledge & authority (who/what expertise is involved; credentials, domain know-how)"

"10. Ethical Axis":
 description: "Ethical considerations (values, moral principles, biases checked)"

"11. Regulatory Axis":
 description: "External laws & regulations (compliance with laws, standards)"

"12. Compliance Axis":
 description: "Internal policy adherence (following internal controls and policies)"

"13. Simulation Axis":
 description: "Hypothetical scenarios & modeling (what-if analysis, simulations run)"

"14. AI-Generated Knowledge Axis":
 description: "AI/Synthetic knowledge (inferences or content produced by AI processes)"

"15. Cultural/Social Axis":
 description: "Cultural and social context (public sentiment, social norms, human factors)"

"16. Innovation/Frontier Axis":
 description: "Innovation & emerging trends (new ideas, novel approaches, research frontiers)"

"17. Learning/Continuous Improvement Axis":
 description: "Meta-learning & self-improvement (how the system learns from experience over time)"

coordinate_parsing:

```

format: "axis_name: value, axis_name: value, ... (17 entries)"
example_input: "Temporal:2025-01-01, Spatial:San Francisco,
Semantic:Healthcare-Diabetes, Procedural:TreatmentPlanA, Analytical:Risk=low,
Normative:Guideline123, Risk:5% complication, Performance:Outcome=Stable,
Expertise:DoctorOpinion, Ethical:ApprovedByIRB, Regulatory:HIPAA,
Compliance:OrgPolicy42, Simulation:What-if-X, AI-Generated:AIInsightID,
Cultural:PatientPref, Innovation:NewTherapy, Learning:UpdatedModel"
example_mapping:
  Temporal: "2025-01-01"
  Spatial: "San Francisco, CA"
  Semantic: "Healthcare > Diabetes"
  Procedural: "TreatmentPlanA protocol"
  Analytical: "Predicted risk = low"
  Normative: "Guideline #123 followed"
  Risk: "5% complication risk"
  Performance: "Outcome stable"
  Expertise: "Doctor's recommendation included"
  Ethical: "IRB approval obtained"
  Regulatory: "HIPAA compliance ensured"
  Compliance: "Org policy 42 satisfied"
  Simulation: "What-if scenario X evaluated"
  AI_Generated: "Insight ID 7 (from AI model)"
  Cultural: "Patient preference accommodated"
  Innovation: "New therapy introduced"
  Learning: "Model updated with outcome data"

```

(Each axis provides a distinct dimension for encoding knowledge, enabling holistic multi-axis representation. The example above shows how a complex event (medical scenario) might be broken down into values along all 17 axes.)

10-Layer Simulation Stack YAML

This YAML defines the 10-layer recursive reasoning stack that the system uses to analyze queries ⁴ ⁵ ⁶. Each layer of the simulation stack is represented with a numbered key, a name, and an explanation of its role. The layers progress from initial context mapping through multi-agent reasoning and final assembly, updating a shared state at each step. Conditional and dynamic behaviors (like adding personas at Layer 4, or performing cross-checks at Layer 8) are built into the design. By Layer 10, a preliminary answer and confidence level are produced, ready for refinement.

```

simulation_stack:
  "Layer 1: Context Initialization":
    purpose: "Map the user query into fundamental axes and determine initial
domain/pillar context"
    actions:
      - "Identify relevant axes from query (e.g. time, location, domain
keywords)"

```

- "Assign initial expert roles or pillars based on query classification"

"Layer 2: Knowledge Expansion":

purpose: "Expand and enrich context by traversing knowledge graph links"

actions:

- "Retrieve related knowledge nodes along each relevant axis (temporal, spatial, etc.)"
- "Broaden context with relevant information branches"

"Layer 3: Primary Agent Activation":

purpose: "Activate primary personas (Knowledge & Sector experts) to analyze the query"

actions:

- "Instantiate Knowledge persona (general expert) and Sector persona (domain-specific expert)"
- "Begin independent analysis of the query by these agents"

"Layer 4: Regulatory & Compliance Integration":

purpose: "Bring in regulatory and compliance checks via specialized personas"

actions:

- "Activate Regulatory persona (external laws/regulations) and Compliance persona (internal policies) ⁷"
- "Inject regulatory/legal context and internal policy constraints into analysis"

"Layer 5: Multi-Agent Orchestration":

purpose: "Orchestrate debate and consensus-building among all active personas"

actions:

- "Enable personas to share findings and raise concerns (e.g. conflicts or risks)"
- "Apply a voting or quorum mechanism to reach an agreed approach"
- "Oversight check: ensure no persona's critical objection is overlooked"

"Layer 6: Pattern Analysis":

purpose: "Perform advanced pattern recognition and deep reasoning across domains"

actions:

- "Analyze aggregated knowledge for patterns or anomalies (e.g. trends, correlations)"
- "Apply machine learning or analytic models to identify insights"
- "Combine cross-domain data to derive higher-level inferences"

"Layer 7: Long-Term Scenario Simulation":

purpose: "Project potential outcomes and long-range implications of proposed solutions"

actions:

- "Simulate scenarios or what-if analyses over an extended timeline ⁸"
- "Evaluate downstream effects or side-effects of actions (risk vs. reward over time)"
- "Incorporate emergent behaviors or future changes into planning"

"Layer 8: Recursive Cross-Verification":

purpose: "Cross-check outputs between personas and against the knowledge

```

graph"
  actions:
    - "Each persona's conclusions are compared for consistency"
    - "Validate answers against known truths in the knowledge base (graph
consistency check) 9 "
    - "Identify any contradictions or dissenting outputs for resolution"
  "Layer 9: Deep Integration":
    purpose: "Ensure holistic coherence by aligning results with the cognitive
model"
    actions:
      - "Integrate multi-axis results into a coherent narrative or solution"
      - "Verify that the solution fits within strategic/long-term objectives"
      - "Finalize any remaining integration of cross-domain knowledge"
  "Layer 10: Emergence & Final Assembly":
    purpose: "Synthesize the preliminary answer and perform final coherence
checks"
    actions:
      - "Aggregate insights from all personas into a draft answer 6 "
      - "Perform 'emergence' check for any subtle contradictions or unintended
effects"
      - "Resolve discrepancies and compute a preliminary confidence score for
the answer"
    shared_state_updates:
      description: "Each layer updates a shared memory state with new findings
and metrics"
      example:
        Layer_5_consensus_strength: 0.81 # e.g., confidence level in consensus
at layer 5
        Layer_8_cross_verification_score: 0.89 # e.g., consistency score among
persona outputs at layer 8
        Layer_10_preliminary_confidence: 0.87 # e.g., confidence score before
refinement

```

(The 10-layer stack progressively builds a comprehensive answer. By Layer 10, the system has integrated inputs from all four personas and performed extensive analysis and cross-checking 9 6 , yielding a well-vetted preliminary result to refine.)

12-Step Refinement Engine YAML

This YAML specifies the 12-step refinement workflow that polishes the simulation output 10 11 . Each step (1 through 12) is listed with its name and a brief description of its function. The refinement engine applies these steps in sequence – from Algorithm of Thought through Final Check – to validate and improve the answer. It also includes a convergence criterion: a confidence threshold (99.5%) that must be met before final output. If the confidence is below the threshold, the engine can loop back and repeat the steps (iterative self-refinement) up to a maximum number of iterations. This ensures the final output achieves the desired confidence level 12 .

```

refinement_engine:
  steps:
    "1. Algorithm of Thought (AoT)":
      description: "Decompose the problem and solution approach into structured sub-tasks"
      related_KA: "KA-001" # Knowledge Agent for AoT (task decomposition)
    "2. Tree of Thought (ToT)":
      description: "Explore multiple reasoning branches in parallel and score their outcomes"
      related_KA: "KA-002"
    "3. Gap Analysis":
      description: "Identify missing knowledge, contradictions or unanswered questions"
      related_KA: "KA-003"
    "4. Data Validation":
      description: "Validate and normalize input data and retrieved evidence for accuracy"
      related_KA: "KA-004"
    "5. NLP Enhancement":
      description: "Improve clarity and coherence of the language in the draft answer (NLP refinement)"
      related_KA: "KA-005"
    "6. Data Consistency Validation":
      description: "Ensure consistency of data/facts across the answer (no internal contradictions)"
      related_KA: "KA-009" # e.g., Evidence Validation KA
    "7. Cross-Reference Validation":
      description: "Cross-check facts and references against external sources or knowledge base"
      related_KA: "KA-009" # (Could reuse Evidence Validation or similar KA)
    "8. Logic Check":
      description: "Verify the reasoning logic and conclusion soundness (no logical fallacies)"
      related_KA: "KA-008" # e.g., Self-Critique & Reflection KA for logic
    "9. Ethical/Bias Audit":
      description: "Audit the answer for ethical issues or bias, and ensure fairness"
      related_KA: "KA-010"
    "10. Regulatory Compliance Check":
      description: "Confirm the answer adheres to all applicable external regulations"
      related_KA: "KA-010" # (Regulatory checks might use a compliance KA as well)
    "11. Security Validation":
      description: "Validate no sensitive data leakage; ensure security and privacy compliance"
      related_KA: "KA-011"

```

```

"12. Final Logical & Compliance Check":
  description: "Perform a final comprehensive check of logic, consistency,
and compliance before output"
  related_KA: "KA-012" # (Final aggregate check, can be conceptual)
  confidence_threshold: 0.995
  max_iterations: 3
  iteration_strategy:
    condition: "confidence < 0.995"
    action: "repeat_steps_1_to_12"
    on_max_iterations: "terminate_with_warning"
  notes:
    - "After step 12, a confidence score is evaluated against the 99.5%
threshold 12 ."
    - "If confidence is below threshold, the engine enters an iterative
refinement loop (up to 3 cycles) to improve the answer."
    - "During iteration, certain persona outputs or weights may be adjusted to
address weaknesses (e.g., boosting Compliance persona influence if a compliance
issue was found)."
    - "If confidence meets threshold or after max iterations, the refined
answer is returned for final output."

```

(The 12-step refinement engine systematically improves the draft answer through multiple quality checks, from logical consistency to ethical compliance ¹⁰ ¹¹ . It uses a high confidence threshold (99.5%) and will iterate the steps up to three times if needed to reach that level of assurance. This ensures the final output is trustworthy and mission-critical quality.)

Quad Persona Templates YAML

This YAML defines the four personas (Knowledge, Sector, Regulatory, Compliance) in a template format. Each persona is described across seven trait dimensions: **role**, **education**, **certifications**, **skills**, **traits**, **job_training**, and **related_roles**. These traits humanize the AI agents, giving them distinct expertise and perspective. Explanation nodes (e.g. `focus`) are included to clarify each persona's primary focus in the simulation ¹ . The personas collaborate during the reasoning process, each bringing their specialty to bear, and the orchestrator mediates their interactions ¹ ⁷ .

```

quad_personas:
  knowledge_persona:
    role: "Knowledge & Reasoning Expert"
    education: "Ph.D. in Information Science (broad domain knowledge)"
    certifications: "Knowledge Engineering Certification"
    skills: "Logical reasoning, cross-domain synthesis, fact-checking"
    traits: "Analytical, objective, detail-oriented"
    job_training: "Trained on comprehensive encyclopedic and scientific corpora"
    related_roles: "Research Analyst, Data Scientist, Librarian"
    focus: "Ensures factual accuracy and logical consistency in answers 1 "
  sector_persona:

```



```

    role: "Domain Specialist Practitioner"
    education: "Master's in Domain-Specific Field (Industry expert)"
    certifications: "Professional Certification in the domain"
    skills: "Deep domain expertise, context-specific decision-making"
    traits: "Pragmatic, goal-oriented, context-aware"
    job_training: "Real-world experience in the sector (industry projects,
field work)"
    related_roles: "Subject Matter Expert, Industry Consultant, Field Operator"
    focus: "Contributes domain-specific context and objectives to ensure
solutions are practical 7 "
    regulatory_persona:
      role: "Regulatory Compliance Advisor"
      education: "J.D. or LLM (Law) with Regulatory focus"
      certifications: "Certified Regulatory Affairs Professional"
      skills: "Legal interpretation, standards compliance, policy mapping"
      traits: "Methodical, risk-averse, principled"
      job_training: "Experience in regulatory agencies or compliance departments"
      related_roles: "Legal Counsel, Regulatory Analyst, Policy Advisor"
      focus: "Ensures alignment with external laws, regulations, and standards 7 "
    compliance_persona:
      role: "Internal Policy & Risk Manager"
      education: "MBA or M.S. in Risk Management/Governance"
      certifications: "Certified Compliance Officer (CCO), Risk Management
Certification"
      skills: "Internal policy enforcement, risk assessment, governance oversight"
      traits: "Cautious, ethical, compliance-focused"
      job_training: "Trained in corporate compliance, audit, and internal
controls"
      related_roles: "Chief Compliance Officer, Risk Manager, Internal Auditor"
      focus: "Enforces internal policies and risk controls to meet organizational
standards 13 "

```

(Each persona template provides the background and perspective of a specialized AI agent in the system. For example, the Knowledge persona is an analytical generalist ensuring factual consistency, whereas the Sector persona brings deep domain insight ¹. The Regulatory and Compliance personas embed external laws and internal policies respectively ⁷. Together, this “Quad” of personas ensures multi-perspective reasoning and balanced, compliant answers.)

Knowledge Agent Registry YAML

This YAML file lists all Knowledge Agents (KAs) in the system from KA-1 to KA-114. Each KA is a modular algorithm or function that can be invoked at specific layers of the simulation or refinement process. The registry provides for each KA: its ID, name, short code, purpose, category, applicable layers, inputs, and outputs. These entries are derived from the KA enterprise registry spreadsheet and define the capabilities available to the UKG system at a granular level. The master orchestration and subsystems call on these KAs as needed (for example, during specific refinement steps or pattern analyses). Below is the complete KA registry in YAML format, capturing the full list of KAs along with their key attributes:

```

knowledge_agents:
  KA-001:
    ka_name: Algorithm of Thought
    short_name: AoT
    purpose: Decompose query into ordered tasks and dependencies
    category: Reasoning
    primary_layers:
      - L1
      - L2
    allowed_layers:
      - L1
      - L2
    inputs: Raw query; policy profile; initial context
    outputs: Task graph (DAG); subtask list
  KA-002:
    ka_name: Tree of Thought
    short_name: ToT
    purpose: Generate and score parallel reasoning branches
    category: Reasoning
    primary_layers:
      - L3
      - L7
    allowed_layers:
      - L3
      - L7
    inputs: Task graph; context state
    outputs: Branch set; branch scores; candidate solutions
  KA-003:
    ka_name: Gap Analysis
    short_name: GAP
    purpose: Detect missing/weak knowledge, contradictions, unknowns
    category: Analysis
    primary_layers:
      - L2
    allowed_layers:
      - L2
    inputs: Context state; evidence set
    outputs: Gap vector; missing-entity list; uncertainty map
  KA-004:
    ka_name: Input Validation & Normalization
    short_name: VALID
    purpose: Sanitize, normalize, and validate user input and metadata
    category: Safety
    primary_layers:
      - L1
    allowed_layers:
      - L1

```

inputs: User input; auth context; policy
 outputs: Normalized query; safe tokens; rejection reasons
 KA-005:
 ka_name: Query Classification
 short_name: QCLASS
 purpose: Classify domain, complexity, stakes, and ambiguity of query
 category: Routing
 primary_layers:
 - L1
 allowed_layers:
 - L1
 inputs: Normalized query; user/org policy
 outputs: Domain tags; complexity tier; risk hints
 KA-006:
 ka_name: Deep Planning
 short_name: PLAN
 purpose: Generate long-chain plan fulfilling goals under constraints
 category: Reasoning
 primary_layers:
 - L7
 allowed_layers:
 - L7
 inputs: Goals; constraints; context
 outputs: Multi-step plan; temporal milestones
 KA-007:
 ka_name: Recursive Reasoning Control
 short_name: RRC
 purpose: Dynamically control recursion depth and breadth in reasoning
 category: Reasoning
 primary_layers:
 - L3
 allowed_layers:
 - L3
 inputs: Branch evaluations; current depth
 outputs: Prune/expand signals; recursion parameters
 KA-008:
 ka_name: Self-Critique & Reflection
 short_name: REFLECT
 purpose: Critique current reasoning path and suggest improvements
 category: Analysis
 primary_layers:
 - L8
 allowed_layers:
 - L8
 inputs: Current solution state; past conclusions
 outputs: Identified flaws; refinement suggestions
 KA-009:
 ka_name: Evidence Validation

short_name: EVALID
 purpose: Cross-verify evidence and claims against trusted sources
 category: Analysis
 primary_layers:
 - L6
 allowed_layers:
 - L6
 inputs: Evidence set; knowledge graph
 outputs: Verified facts; invalid evidence flags
 KA-010:
 ka_name: Bias Detection
 short_name: BIAS
 purpose: Identify potential biases or ethical issues in content
 category: Safety
 primary_layers:
 - L9
 - L10
 allowed_layers:
 - L9
 - L10
 inputs: Draft answer; context
 outputs: Bias/ethics report; mitigation suggestions
 KA-011:
 ka_name: Analytical Modeling
 short_name: MODEL
 purpose: Apply statistical or ML models to derive insights (predictive analytics)
 category: Analysis
 primary_layers:
 - L6
 allowed_layers:
 - L6
 inputs: Processed data; features
 outputs: Model results; forecasts; anomaly detections
 KA-012:
 ka_name: Outcome Synthesis
 short_name: SYNTH
 purpose: Synthesize multi-axis outcomes into a unified recommendation
 category: Synthesis
 primary_layers:
 - L10
 allowed_layers:
 - L10
 inputs: Multi-axis intermediate results
 outputs: Unified solution draft; executive summary
 KA-013:
 ka_name: AI-Generated Knowledge Integrator
 short_name: AI-KNOW

```

    purpose: Integrate AI inferences (from AI knowledge axis) into the evidence
base
    category: Knowledge
    primary_layers:
      - L4
      - L6
    allowed_layers:
      - L4
      - L6
    inputs: AI-generated insights; baseline knowledge
    outputs: Augmented knowledge graph; AI inference log
KA-014:
    ka_name: Cultural Context Adapter
    short_name: CULTURE
    purpose: Adjust recommendations based on cultural and social context
    category: Adaptation
    primary_layers:
      - L5
    allowed_layers:
      - L5
    inputs: Draft recommendations; cultural norms database
    outputs: Culturally adapted advice; sensitivity flags
KA-015:
    ka_name: Innovation Tracker
    short_name: INNOV
    purpose: Incorporate latest research or innovative methods relevant to the
problem
    category: Research
    primary_layers:
      - L7
    allowed_layers:
      - L7
    inputs: Domain context; innovation database
    outputs: Suggested new approaches; emerging trend info
KA-016:
    ka_name: Strategic Alignment Checker
    short_name: STRAT
    purpose: Ensure recommendations align with long-term strategic goals
    category: Governance
    primary_layers:
      - L9
    allowed_layers:
      - L9
    inputs: Proposed solution; strategic objectives
    outputs: Alignment score; strategic fit notes
KA-017:
    ka_name: Continuous Learning Updater
    short_name: LEARN

```

purpose: Update knowledge base with newly learned outcomes and feedback
category: Lifecycle
primary_layers:
- L10
allowed_layers:
- L10
inputs: Outcome data; feedback signals
outputs: Knowledge graph updates; model adjustment signals

KA-018:

ka_name: Temporal Trend Analyzer
short_name: TRENDS
purpose: Analyze time-series data for trends and temporal patterns
category: Analysis
primary_layers:
- L5
- L6
allowed_layers:
- L5
- L6
inputs: Temporal data series; events timeline
outputs: Trend analysis report; forecasted timeline events

KA-019:

ka_name: Spatial Impact Mapper
short_name: MAP
purpose: Evaluate geographic/spatial impact and correlations
category: Analysis
primary_layers:
- L5
allowed_layers:
- L5
inputs: Geospatial data; location tags
outputs: Impact map; location-based risk factors

KA-020:

ka_name: Ontology Alignment Engine
short_name: ONTALIGN
purpose: Align and merge concepts across different ontologies (semantic

axis)

category: Knowledge
primary_layers:
- L3
- L6
allowed_layers:
- L3
- L6
inputs: Semantic nodes; ontology mappings
outputs: Unified concept mappings; ontology alignment report

KA-021:

ka_name: Emergence Detection

```

short_name: EMERG
purpose: Detect novel patterns/insights; flag unexpected behavior
category: Analysis
primary_layers:
  - L7
  - L10
allowed_layers:
  - L7
  - L10
inputs: Reasoning graph; agent behaviors
outputs: Emergence flags; novelty score
KA-022:
  ka_name: Risk Assessment
  short_name: RISK
  purpose: Compute operational/mission risk of recommendations
  category: Safety
  primary_layers:
    - L8
    - L9
  allowed_layers:
    - L8
    - L9
  inputs: Consolidated findings; scenario data
  outputs: Risk score; risk breakdown
KA-023:
  ka_name: Outcome Confidence Calibrator
  short_name: CALIB
  purpose: Calibrate confidence scores across subsystems for consistency
  category: Truth
  primary_layers:
    - L10
  allowed_layers:
    - L10
  inputs: Persona confidence scores; validation results
  outputs: Unified confidence score; calibration report
KA-024:
  ka_name: Red Team Simulator
  short_name: REDTEAM
  purpose: Simulate adversarial perspective to test robustness of solutions
  category: Safety
  primary_layers:
    - L7
  allowed_layers:
    - L7
  inputs: Proposed solution; threat models
  outputs: Identified weaknesses; stress test results
KA-025:
  ka_name: Blue Team Validator

```

```

    short_name: BLUETEAM
    purpose: Simulate supportive/use-case perspective to validate solution
utility
    category: Safety
    primary_layers:
      - L7
    allowed_layers:
      - L7
    inputs: Proposed solution; expected benefits
    outputs: Utility confirmation; scenarios of success
KA-026:
    ka_name: Persona Quorum Evaluator
    short_name: QUORUM
    purpose: Evaluate if personas' outputs meet quorum/consensus thresholds
    category: Governance
    primary_layers:
      - L5
    allowed_layers:
      - L5
    inputs: Persona recommendations; weights
    outputs: Consensus score; dissent details
KA-027:
    ka_name: Ethical Framework Mapper
    short_name: ETHICS
    purpose: Map outputs to ethical frameworks/principles for evaluation
    category: Safety
    primary_layers:
      - L9
    allowed_layers:
      - L9
    inputs: Draft output; ethical guidelines
    outputs: Ethical compliance report; flagged issues
KA-028:
    ka_name: GDPR/PII Scrubber
    short_name: PII
    purpose: Remove or anonymize personal identifiable info and sensitive data
    category: Safety
    primary_layers:
      - L11
    allowed_layers:
      - L11
    inputs: Draft output; compliance rules
    outputs: Sanitized output; data handling log
KA-029:
    ka_name: Knowledge Gaps Filler
    short_name: FILLER
    purpose: Attempt to fill small knowledge gaps using background knowledge
    category: Knowledge

```



```

primary_layers:
  - L2
allowed_layers:
  - L2
inputs: Gap list; UKG knowledge base
outputs: Filled facts; notes on assumptions
KA-030:
  ka_name: Hypothesis Generator
  short_name: HYPOTH
  purpose: Generate plausible hypotheses for unexplained aspects of the query
  category: Reasoning
  primary_layers:
    - L3
  allowed_layers:
    - L3
  inputs: Gap analysis; context
  outputs: Hypothesis list; associated confidence
KA-031:
  ka_name: Plan Variance Analyzer
  short_name: PLANVAR
  purpose: Analyze differences between possible plans/strategies
  category: Analysis
  primary_layers:
    - L7
  allowed_layers:
    - L7
  inputs: Multiple plans; objectives
  outputs: Comparison report; variance metrics
KA-032:
  ka_name: Socioeconomic Impact Assessor
  short_name: SOCIO
  purpose: Assess social and economic impacts of recommendations
  category: Analysis
  primary_layers:
    - L8
  allowed_layers:
    - L8
    - L9
  inputs: Proposed solution; demographic/economic data
  outputs: Impact assessment; equity considerations
KA-033:
  ka_name: Timeline Consistency Checker
  short_name: TIMECHK
  purpose: Ensure temporal consistency of events and schedules
  category: Analysis
  primary_layers:
    - L8
  allowed_layers:

```

- L8

inputs: Proposed timeline; known events
outputs: Consistency report; adjusted timeline

KA-034:

ka_name: Spatial Consistency Checker
short_name: SPACECHK
purpose: Ensure spatial/geographic consistency and feasibility
category: Analysis
primary_layers:

- L8

allowed_layers:

- L8

inputs: Locations involved; logistical data
outputs: Feasibility report; map consistency notes

KA-035:

ka_name: Query Decomposer
short_name: QDECOMP
purpose: Break down complex queries into simpler sub-queries
category: Reasoning
primary_layers:

- L1

allowed_layers:

- L1

inputs: Raw query (complex/multi-part)
outputs: Sub-query list; dependency graph

KA-036:

ka_name: Response Synthesizer
short_name: SYNTHRESP
purpose: Synthesize responses from multiple personas into one coherent answer

category: Synthesis
primary_layers:

- L10

allowed_layers:

- L10

inputs: Persona outputs; answer outline
outputs: Unified draft answer; rationale mapping

KA-037:

ka_name: Conflict Resolver
short_name: RESOLVE
purpose: Resolve conflicts between persona recommendations
category: Governance
primary_layers:

- L5

allowed_layers:

- L5

inputs: Conflicting points; persona justifications
outputs: Compromise solution; conflict resolution log

KA-038:

ka_name: Context Scope Expander

short_name: EXPAND

purpose: Broaden or narrow the context scope as needed for thoroughness

category: Adaptation

primary_layers:

- L2

allowed_layers:

- L2

inputs: Initial context; query classification

outputs: Adjusted context scope; additional context data

KA-039:

ka_name: Memory Compaction & Recall

short_name: MEMORY

purpose:

Retrieve relevant info from long-term memory and compact irrelevant details

category: Knowledge

primary_layers:

- L2

- L6

allowed_layers:

- L2

- L6

inputs: Initial knowledge state; query context

outputs: Focused knowledge subset; memory recall log

KA-040:

ka_name: Provenance Tracker

short_name: PROV

purpose: Track sources and provenance of each fact or statement

category: Truth

primary_layers:

- L6

- L10

allowed_layers:

- L6

- L10

inputs: Knowledge graph edges; external references

outputs: Provenance map; confidence adjustments

KA-041:

ka_name: Concept Confidence Normalization

short_name: CNORM

purpose: Normalize confidence scales across domains

category: Truth

primary_layers:

- L6

allowed_layers:

- L6

inputs: Confidence vectors

outputs: Normalized confidence
 KA-042:
 ka_name: Contradiction Propagation Analysis
 short_name: CPROP
 purpose: Trace downstream impacts of contradictions
 category: Analysis
 primary_layers:
 - L6
 allowed_layers:
 - L6
 inputs: Identified contradiction; knowledge links
 outputs: Affected nodes list; inconsistency impact report
 KA-043:
 ka_name: Query Ambiguity Resolver
 short_name: QAMB
 purpose: Clarify ambiguous query elements (ask clarifying questions if needed)
 category: Reasoning
 primary_layers:
 - L1
 allowed_layers:
 - L1
 inputs: Raw query; ambiguity flags
 outputs: Disambiguated query; assumptions made
 KA-044:
 ka_name: Multi-Modal Data Interpreter
 short_name: MULTI
 purpose: Interpret non-text data (images, charts) referenced in query
 category: Capability
 primary_layers:
 - L3
 allowed_layers:
 - L3
 inputs: Embedded image or data; context
 outputs: Textual description; integrated data points
 KA-045:
 ka_name: Staleness Checker
 short_name: STALE
 purpose: Check if information might be outdated and needs refresh
 category: Lifecycle
 primary_layers:
 - L6
 allowed_layers:
 - L6
 inputs: Knowledge timestamp; current date
 outputs: Staleness alert; refresh suggestions
 KA-046:
 ka_name: Model Confidence Adjuster

short_name: MCADJ
 purpose: Adjust persona confidences based on past performance and bias
 category: Adaptation
 primary_layers:
 - L9
 allowed_layers:
 - L9
 inputs: Persona track record; outcome logs
 outputs: Adjusted persona weightings; confidence modifiers

KA-047:

ka_name: Federated Data Fetcher
 short_name: FEDFETCH
 purpose: Retrieve data from federated sources without central storage
 category: Capability
 primary_layers:
 - L3
 allowed_layers:
 - L3
 inputs: Sub-query; source metadata
 outputs: Retrieved data; source references

KA-048:

ka_name: Persona Hand-off Manager
 short_name: HANDOFF
 purpose: Manage passing of context between personas across layers
 category: Governance
 primary_layers:
 - L5
 allowed_layers:
 - L5
 inputs: Persona outputs; context snapshot
 outputs: Handoff instructions; context alignment report

KA-049:

ka_name: Anomaly Detector
 short_name: ANOM
 purpose: Detect anomalies or outliers in data and reasoning
 category: Analysis
 primary_layers:
 - L6
 allowed_layers:
 - L6
 inputs: Analytical data; expected patterns
 outputs: Anomaly report; flagged items

KA-050:

ka_name: Performance Optimizer
 short_name: OPTIM
 purpose: Optimize performance metrics (speed, efficiency) in solution
 category: Optimization
 primary_layers:

- L8

allowed_layers:

- L8

inputs: Current solution plan; performance metrics
outputs: Optimization suggestions; streamlined plan

KA-051:

ka_name: Cost-Benefit Analyzer
short_name: COSTBEN
purpose: Analyze cost vs benefit for proposed plans
category: Analysis
primary_layers:

- L7

allowed_layers:

- L7

inputs: Plan outline; cost metrics; benefit metrics
outputs: Cost-benefit analysis report; recommendation

KA-052:

ka_name: Sentiment Analyzer
short_name: SENTI
purpose: Gauge public or stakeholder sentiment related to the query
category: Analysis
primary_layers:

- L5

allowed_layers:

- L5

inputs: Social data; feedback corpus
outputs: Sentiment score; prevailing opinion summary

KA-053:

ka_name: Persona Bias Calibrator
short_name: PBias
purpose: Detect and correct any one persona dominating or biasing results
category: Governance
primary_layers:

- L5

allowed_layers:

- L5

inputs: Persona outputs; weights
outputs: Rebalanced weights; bias report

KA-054:

ka_name: Knowledge Base Curation
short_name: CURATE
purpose: Curate relevant knowledge from UKG for this query (scoping)
category: Knowledge
primary_layers:

- L2

allowed_layers:

- L2

inputs: Full knowledge graph; query context

outputs: Focused knowledge subset; curation log

KA-055:

ka_name: Explanability Tracer

short_name: EXPLAIN

purpose: Trace reasoning steps to provide explanation for the answer

category: Truth

primary_layers:

- L10

allowed_layers:

- L10

inputs: Reasoning graph; decision points

outputs: Explanation report; reasoning trace

KA-056:

ka_name: Defensive Argumentation

short_name: DEFEND

purpose: Prepare counter-arguments to anticipated objections

category: Reasoning

primary_layers:

- L8

allowed_layers:

- L8

inputs: Proposed answer; known objections

outputs: Rebuttal points; confidence in defense

KA-057:

ka_name: External Knowledge Integrator

short_name: EXTKNOW

purpose: Integrate knowledge from external databases or APIs

category: Capability

primary_layers:

- L3

allowed_layers:

- L3

inputs: API results; external DB query

outputs: Integrated data nodes; source citations

KA-058:

ka_name: Internal Consistency Enforcer

short_name: CONSIST

purpose: Enforce consistency of answers with internal historical decisions

category: Governance

primary_layers:

- L9

allowed_layers:

- L9

inputs: Proposed answer; past decisions log

outputs: Consistency approval or adjustments; justification log

KA-059:

ka_name: Role Context Switcher

short_name: ROLESWAP

purpose: Switch perspective to another persona's role if needed (for "what would X say" questions)
category: Adaptation
primary_layers:
- L5
allowed_layers:
- L5
inputs: Current persona outputs; target perspective
outputs: Perspective-shifted advice; comparison report

KA-060:

ka_name: Meta-Reasoning Monitor
short_name: META
purpose: Monitor the reasoning process itself for inefficiencies or loops
category: Governance
primary_layers:
- L8
allowed_layers:
- L8
inputs: Reasoning process trace; intermediate states
outputs: Meta-level observations; optimization suggestions

KA-061:

ka_name: Adversarial Input Shield
short_name: SHIELD
purpose: Detect/neutralize malicious inputs
category: Safety
primary_layers:
- L1
allowed_layers:
- L1
inputs: Raw query; metadata
outputs: Sanitized query; block decision

KA-062:

ka_name: Decentralized Trust Scoring
short_name: D-TRUST
purpose: Compute trust score via provenance hashes/ledger concepts
category: Truth
primary_layers:
- L6
- L10
allowed_layers:
- L6
- L10
inputs: Source provenance data; ledger records
outputs: Trust score; trust breakdown

KA-063:

ka_name: Data Imputation Engine
short_name: IMPUTE
purpose: Impute or infer missing data points from context


```

category: Analysis
primary_layers:
  - L2
  - L6
allowed_layers:
  - L2
  - L6
inputs: Dataset with gaps; statistical models
outputs: Imputed values; confidence intervals
KA-064:
ka_name: Multi-Lingual Interpreter
short_name: LANG
purpose: Handle queries or data in multiple languages
category: Capability
primary_layers:
  - L1
  - L3
allowed_layers:
  - L1
  - L3
inputs: Non-English query/data; translation models
outputs: Translated query; normalized context
KA-065:
ka_name: Knowledge Graph Expander
short_name: GRAPHX
purpose: Expand the UKG with new nodes/edges discovered during reasoning
category: Knowledge
primary_layers:
  - L6
  - L10
allowed_layers:
  - L6
  - L10
inputs: New insights; unlinked data
outputs: Updated knowledge graph; expansion log
KA-066:
ka_name: Compliance Mesh Mapper
short_name: MESH
purpose: Map regulatory frameworks to internal controls (compliance mesh)
category: Governance
primary_layers:
  - L4
  - L9
allowed_layers:
  - L4
  - L9
inputs: Compliance frameworks; internal policy graph
outputs: Mapped control requirements; compliance checklist

```

KA-067:

ka_name: Privacy Checker
short_name: PRIVACY
purpose: Check answers for privacy compliance (GDPR, HIPAA, etc.)
category: Safety
primary_layers:
- L11
allowed_layers:
- L11
inputs: Draft answer; privacy rules
outputs: Privacy compliance report; redaction suggestions

KA-068:

ka_name: Effort Estimator
short_name: EFFORT
purpose: Estimate effort or resources required for proposed solutions
category: Analysis
primary_layers:
- L7
allowed_layers:
- L7
inputs: Solution plan; resource models
outputs: Effort estimate; resource plan

KA-069:

ka_name: Reliability Assessor
short_name: RELIAB
purpose: Assess reliability and robustness of the solution
category: Analysis
primary_layers:
- L9
allowed_layers:
- L9
inputs: Solution components; reliability data
outputs: Reliability score; single point of failure analysis

KA-070:

ka_name: Feedback Loop Inserter
short_name: FEED
purpose: Insert feedback mechanisms for continuous learning
category: Lifecycle
primary_layers:
- L10
allowed_layers:
- L10
inputs: Outcome metrics; user feedback
outputs: Feedback prompts; model update signals

KA-071:

ka_name: Context Re-Evaluator
short_name: REEVAL
purpose: Re-evaluate context if the solution seems off-base (triggers

```

partial restart)
  category: Adaptation
  primary_layers:
    - L8
  allowed_layers:
    - L8
  inputs: Near-final answer; context mismatch signals
  outputs: Context adjustment recommendation; restart flag
KA-072:
  ka_name: Constraint Solver
  short_name: SOLVE
  purpose: Solve for optimal solution given a set of constraints
  category: Reasoning
  primary_layers:
    - L7
  allowed_layers:
    - L7
  inputs: Constraints; objective function
  outputs: Optimal solution parameters; feasibility report
KA-073:
  ka_name: Unit Test Generator
  short_name: UTEST
  purpose: Generate test cases or scenarios to validate the answer
  category: Safety
  primary_layers:
    - L9
  allowed_layers:
    - L9
  inputs: Final answer draft; critical assumptions
  outputs: Test scenarios; expected outcomes
KA-074:
  ka_name: Historical Correlation Analyzer
  short_name: HISTCORR
  purpose: Correlate with historical cases or data for validation
  category: Analysis
  primary_layers:
    - L6
  allowed_layers:
    - L6
  inputs: Current case data; historical dataset
  outputs: Correlation findings; precedent cases
KA-075:
  ka_name: Domain Ontology Updater
  short_name: ONTUPDATE
  purpose: Update domain ontology with new concepts learned
  category: Knowledge
  primary_layers:
    - L10

```

```

    allowed_layers:
      - L10
    inputs: New concept definitions; ontology
    outputs: Updated ontology; change log
KA-076:
  ka_name: Anonymizer
  short_name: ANON
  purpose: Ensure output is anonymized where required (related to privacy)
  category: Safety
  primary_layers:
    - L11
  allowed_layers:
    - L11
  inputs: Draft output; PII markers
  outputs: Anonymized output; anonymization report
KA-077:
  ka_name: Multi-Objective Optimizer
  short_name: MOPT
  purpose: Optimize solution considering multiple objectives (trade-offs)
  category: Optimization
  primary_layers:
    - L7
  allowed_layers:
    - L7
  inputs: Objectives; weights; solution space
  outputs: Pareto-optimal solutions; trade-off analysis
KA-078:
  ka_name: Data Integrity Verifier
  short_name: DIV
  purpose: Verify that data has not been tampered with (integrity check)
  category: Safety
  primary_layers:
    - L6
  allowed_layers:
    - L6
  inputs: Data inputs; hash/signature
  outputs: Integrity verification result; alerts
KA-079:
  ka_name: Redundancy Checker
  short_name: REDUND
  purpose: Check for redundant steps or information in the solution
  category: Optimization
  primary_layers:
    - L8
  allowed_layers:
    - L8
  inputs: Solution plan; answer content
  outputs: Redundancy report; simplification suggestions

```

KA-080:

ka_name: Model Escalation Handler
short_name: ESCALATE
purpose: Decide when to escalate to a larger model or external tool
category: Capability
primary_layers:

- L3

allowed_layers:

- L3

inputs: Query complexity; confidence shortfall
outputs: Escalation decision; external call trigger

KA-081:

ka_name: Simulation Budget Enforcer
short_name: BUDGET
purpose: Enforce compute/recursion limits
category: Safety
primary_layers:

- L7

allowed_layers:

- L7

inputs: Cost estimate; policy caps
outputs: Allow/restrict; new caps

KA-082:

ka_name: Confidence Drift Monitor
short_name: DRIFT
purpose: Detect gradual confidence degradation
category: Lifecycle
primary_layers:

- L10

allowed_layers:

- L10

inputs: Confidence history; recent outcomes
outputs: Drift alert; recalibration suggestion

KA-083:

ka_name: Data Pipeline Auditor
short_name: DPAudit
purpose: Audit data pipeline for errors or inconsistencies
category: Safety
primary_layers:

- L6

allowed_layers:

- L6

inputs: Data pipeline logs; outputs
outputs: Audit report; anomaly flags

KA-084:

ka_name: Compliance Change Notifier
short_name: COMPNOTIFY
purpose: Alert system to new or changed regulations/policies

```

category: Governance
primary_layers:
  - L4
  - L11
allowed_layers:
  - L4
  - L11
inputs: Regulatory feed; policy updates
outputs: Update alerts; policy change log
KA-085:
  ka_name: Environment Context Modeler
  short_name: ENVCON
  purpose: Incorporate environmental context (season, weather, etc.) if
relevant
  category: Knowledge
  primary_layers:
    - L2
  allowed_layers:
    - L2
  inputs: Environment data; query context
  outputs: Context annotations; environment factors
KA-086:
  ka_name: Simulation Outcome Predictor
  short_name: PREDICT
  purpose:
Predict outcomes of simulation without fully running (short-circuit known
results)
  category: Optimization
  primary_layers:
    - L7
  allowed_layers:
    - L7
  inputs: Scenario parameters; historical outcomes
  outputs: Outcome prediction; confidence
KA-087:
  ka_name: Inter-Persona Consistency Checker
  short_name: IPCONSIST
  purpose: Check consistency between persona perspectives in final answer
  category: Governance
  primary_layers:
    - L9
  allowed_layers:
    - L9
  inputs: Persona outputs; final draft
  outputs: Consistency report; sections to reconcile
KA-088:
  ka_name: Monte Carlo Simulator
  short_name: MONTE

```

purpose: Run Monte Carlo simulations for probabilistic analysis
 category: Capability
 primary_layers:
 - L7
 allowed_layers:
 - L7
 inputs: Simulation model; stochastic variables
 outputs: Distribution of outcomes; risk profile

KA-089:

ka_name: Neural Pattern Recognizer
 short_name: NEURAL
 purpose: Use neural networks to recognize complex patterns in data
 category: Capability
 primary_layers:
 - L6
 allowed_layers:
 - L6
 inputs: Processed data; neural model
 outputs: Pattern findings; learned features

KA-090:

ka_name: Explanation Validator
 short_name: EXPVAL
 purpose: Validate that explanations provided are correct and understandable
 category: Truth
 primary_layers:
 - L10
 allowed_layers:
 - L10
 inputs: Explanation report; user profile
 outputs: Clarity score; accuracy check

KA-091:

ka_name: Parallelism Manager
 short_name: PARALLEL
 purpose: Manage parallel execution of tasks or reasoning branches
 category: Optimization
 primary_layers:
 - L3
 allowed_layers:
 - L3
 inputs: Task graph; resource status
 outputs: Parallel execution plan; sync points

KA-092:

ka_name: Emotional Tone Assessor
 short_name: TONE
 purpose: Assess and adjust emotional tone of the answer (for user reception)
 category: Adaptation
 primary_layers:
 - L10

allowed_layers:
 - L10
 inputs: Draft answer; user profile
 outputs: Tone metrics; adjustment suggestions

KA-093:

ka_name: Value Alignment Checker
 short_name: ALIGN
 purpose: Ensure answer aligns with stated values/principles of organization
 category: Governance
 primary_layers:
 - L9
 allowed_layers:
 - L9
 inputs: Final draft; organization values
 outputs: Alignment report; value conflict flags

KA-094:

ka_name: Failure Mode Analyzer
 short_name: FAILSAFE
 purpose: Analyze potential failure modes of proposed solution
 category: Safety
 primary_layers:
 - L8
 allowed_layers:
 - L8
 inputs: Solution description; risk analysis
 outputs: Failure mode list; mitigation strategies

KA-095:

ka_name: Incentive Evaluator
 short_name: INCENT
 purpose: Evaluate incentives of stakeholders to identify misalignment
 category: Analysis
 primary_layers:
 - L9
 allowed_layers:
 - L9
 inputs: Stakeholder list; outcome impacts
 outputs: Incentive alignment report; conflict areas

KA-096:

ka_name: Secure Execution Monitor
 short_name: SECUREX
 purpose: Monitor that simulation execution adheres to security constraints
 category: Safety
 primary_layers:
 - L6
 allowed_layers:
 - L6
 inputs: Execution logs; security policies
 outputs: Security compliance status; alerts

KA-097:

ka_name: API Call Manager
short_name: APIMGR
purpose: Safely manage external API calls and integrate results
category: Capability
primary_layers:

- L3

allowed_layers:

- L3

inputs: API query; policy (allowed calls)
outputs: API results; sanitized data

KA-098:

ka_name: Sandbox Environment Manager
short_name: SANDBOX
purpose: Isolate and test risky operations in a sandbox environment
category: Safety
primary_layers:

- L7

allowed_layers:

- L7

inputs: Operation specs; sandbox policy
outputs: Sandbox results; safety report

KA-099:

ka_name: Outcome Reviewer
short_name: REVIEW
purpose: Summarize and review the outcome against initial query and objectives
category: Synthesis
primary_layers:

- L10

allowed_layers:

- L10

inputs: Final answer; initial query; objectives
outputs: Review report; verification checklist

KA-100:

ka_name: Memory Snapshotter
short_name: SNAPSHOT
purpose: Take snapshot of system state at key points for rollback/analysis
category: Lifecycle
primary_layers:

- L7
- L10

allowed_layers:

- L7
- L10

inputs: Trigger event; current state
outputs: State snapshot; state ID

KA-101:

```

ka_name: Chaos Injection Governor
short_name: CHAOS-GOV
purpose: Policy control for chaos injection magnitude and scope
category: Control
primary_layers:
  - L3
  - L4
  - L5
  - L6
  - L7
allowed_layers:
  - L3
  - L4
  - L5
  - L6
  - L7
inputs: Entropy; risk; mode
outputs: Chaos bounds; allowed KAs
KA-102:
  ka_name: Global Entropy Quantifier
  short_name: ENTROPY
  purpose: Compute normalized entropy score for simulation
  category: Control
  primary_layers:
    - L6
  allowed_layers:
    - L6
  inputs: Branch variance; persona variance; evidence variance
  outputs: Entropy score [0..1]; components
KA-103:
  ka_name: Simulation State Rollback Manager
  short_name: ROLLBACK
  purpose: Checkpoint and rollback simulation state
  category: Safety
  primary_layers:
    - L7
  allowed_layers:
    - L7
  inputs: Checkpoint state; failure flags
  outputs: Restored state; rollback log
KA-104:
  ka_name: Truth vs Utility Arbiter
  short_name: TUA
  purpose: Balance correctness vs operational safety/utility
  category: Safety
  primary_layers:
    - L8
  allowed_layers:

```

- L8

inputs: Truth scores; risk profile; audience
outputs: Adjusted framing; veto if harmful

KA-105:

ka_name: Conceptual Obsolescence Monitor
short_name: OBSOLETE
purpose: Detect paradigm-level obsolescence (not just staleness)
category: Lifecycle
primary_layers:

- L10

allowed_layers:

- L10

inputs: Contradiction trends; policy shifts; citations
outputs: Obsolescence flags; revalidation requests

KA-106:

ka_name: Human Override Reason Capture
short_name: OVERRIDE
purpose: Capture human correction rationale in structured form
category: Governance
primary_layers:

- L10

allowed_layers:

- L10

inputs: Human decision; rationale
outputs: Structured correction; training signal

KA-107:

ka_name: Reasoning Boundary Enforcer
short_name: BOUNDARY
purpose: Enforce hard boundaries on capabilities and layer entry
category: Safety
primary_layers:

- L1
- L7

allowed_layers:

- L1
- L7

inputs: Policy; query class; planned pipeline
outputs: Allowed capabilities; blocks

KA-108:

ka_name: Capability Escalation Detector
short_name: CAP-ESC
purpose: Detect unsafe capability drift/escalation patterns
category: Safety
primary_layers:

- L6
- L10

allowed_layers:

- L6

- L10

inputs: KA interaction patterns; emergence flags
outputs: Escalation alerts; containment actions

KA-109:

ka_name: Knowledge Containment Classifier
short_name: CONTAIN
purpose: Classify knowledge persistence safety (public/restricted/never)
category: Governance
primary_layers:

- L10

allowed_layers:

- L10

inputs: Knowledge candidate; risk signals
outputs: Containment class; persistence rules

KA-110:

ka_name: Cross-Domain Coupling Risk Analyzer
short_name: COUPLE-RISK
purpose: Detect risky cross-domain coupling pathways
category: Safety
primary_layers:

- L6
- L7

allowed_layers:

- L6
- L7

inputs: Domain tags; planned KAs; knowledge links
outputs: Coupling risk score; blocked combos

KA-111:

ka_name: Long-Horizon Goal Drift Monitor
short_name: GOAL-DRIFT
purpose: Detect latent goal persistence across runs
category: Safety
primary_layers:

- L10

allowed_layers:

- L10

inputs: Evolution actions; traces
outputs: Goal-drift alerts; constraints

KA-112:

ka_name: System Self-Introspection Auditor
short_name: INTROSPECT
purpose: 'Audit system-level behavior: chaos usage, overrides, drift'
category: Governance
primary_layers:

- L10

allowed_layers:

- L10

inputs: System metrics; logs

```

    outputs: Audit report; governance actions
KA-113:
  ka_name: Query Analysis & Complexity Router
  short_name: QROUTER
  purpose: Select how much of the system runs based on difficulty/stakes
  category: Routing
  primary_layers:
    - L1
  allowed_layers:
    - L1
  inputs: Normalized query; org mode; policy
  outputs: Complexity tier; layer depth; budgets
KA-114:
  ka_name: External Deep Research Invocation
  short_name: DEEP-EXT
  purpose: Invoke bounded Deep Research provider and return evidence packet
  category: Capability
  primary_layers:
    - L3
  allowed_layers:
    - L3
  inputs: Sub-question; scope; timebox; policy
  outputs: Research packet; source list; citations

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(The Knowledge Agent registry above enumerates all KAs (1–114) with their definitions. These range from reasoning algorithms like Algorithm of Thought (KA-001) and Tree of Thought (KA-002), through validation and safety checks (e.g. Bias Detection KA-010, Adversarial Shield KA-061), up to orchestration and meta-level controllers (e.g. Chaos Injection Governor KA-101). Each entry lists where in the 10-layer/12-step processes it operates (e.g. primary/allowed layers) and what inputs/outputs it expects. This library of KAs is plugged into the simulation and refinement workflow, enabling the system’s rich functionality.)

Sources: The 17-axis schema and multi-layer architecture are based on the UKG technical architecture ³ ⁴. The quad-persona framework and their focuses on factual, domain, regulatory, and compliance perspectives are drawn from the UKG documentation ¹. The 10-layer reasoning stack and 12-step refinement workflow are built as specified in the UKG design docs ⁴ ¹⁰. The Knowledge Agent registry (KA-1 to KA-114) is adapted from the provided enterprise registry spreadsheet, ensuring naming and functionality consistent with the UKG framework. All YAML configurations above collectively serve as the execution blueprint for the full 17-axis UKG/USKD system, integrating knowledge representation, multi-agent reasoning, iterative refinement, and governance controls into a unified, interpretable format.

¹ ² ³ ⁷ ¹³ The_17_Axis_System__Comprehensive_Technical_Architecture.pdf

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⁴ ⁵ ⁶ ⁸ ⁹

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